

Sandbox AI: We Don't Trust Each Other but Want to Create New Value Efficiently Through Collaboration Using Sensitive Data

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ABSTRACT

This research deals with how to build reliable AI models using shared sensitive data. Confidential computing is gaining attention in AI services for a ubiquitous computing field. It makes legal and social regulations strict, including protecting privacy and ensuring security for sensitive data and AI models. As is also the case in ubiquitous computing research. The research phase also requires a manual trial-and-error process to create value through new combinations of shared data and AI models. Conventional confidential computing cannot realize the human workflow. This paper proposes and verifies "Sandbox AI" that can build precise AI models through efficient manual processes without revealing the shared data or models to one another. Sandbox AI utilizes privacy-preserving synthetic data generation and active learning on a confidential computing architecture. Specifically, Sandbox AI works in a sandbox container and generates synthetic data on the basis of real data, interactively extracts synthetic data that are useful for efficient model training while considering the real data distribution, discloses only the extracted synthetic data for annotation, and re-trains the model on the labeled synthetic data. Experimental results show that the achieved model accuracy is as good as that of conventional learning using real data with privacy violations.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools; • Security and privacy → Usability in security and privacy.

KEYWORDS

privacy-preserving data synthesis, active learning, sandbox, human-in-the-loop machine learning, confidential computing

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1 INTRODUCTION

A long time has passed since AI analysis combining personal wearable data and medical data in hospitals first gained popularity. However, for such data and AI models, healthcare services require the confidentiality, integrity, and availability of information under stricter legal and social restrictions on privacy protection and security assurance [13]. The leading approach to meet the requirements is confidential computing [28]. Confidential computing mainly consists of an isolated execution environment architected in an encrypted memory region and remote attestation [7].

Ubiquitous computing studies also have to satisfy similar demands. They additionally need to consider complex human actors. For example, patients provide wearable data, hospitals provide medical data, domain experts annotate the shared data, data scientists build and train AI models, and users use the output of the models. Conventional confidential computing can only provide outputs of static algorithms for shared data. It cannot handle annotation or trial-and-error model building processes for ubiquitous computing research.

Privacy-preserving synthetic data generation is a promising approach to fill the gaps. It is a technology to generate new data with similar properties and differential privacy (DP) [10] derived from real data [20]. Applying privacy-preserving synthetic data to training data instead of real data can reduce privacy risks and accelerate sensitive data sharing by achieving both privacy preservation and data usability [6, 12]. However, existing works focus only on the generation of synthetic data including label information. No framework has been explored for AI model building, including annotation, in a confidential computing architecture.

This research goal is to build precise AI models through efficient manual processes without revealing any shared sensitive data or AI models to one another. Specifically, our framework works in a sandbox container and generates privacy-preserving synthetic

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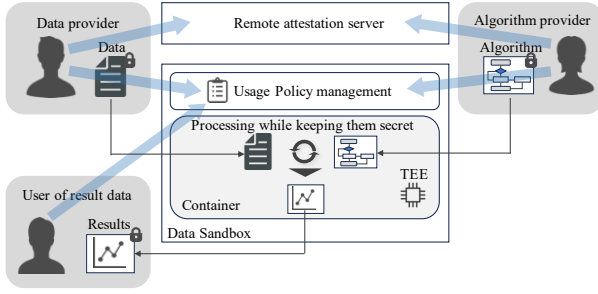


Figure 1: Data sandbox

data on the basis of real data, interactively extracts synthetic data that are useful for efficient model training by active learning while considering the distribution of real data, discloses only the extracted synthetic data for annotation, and re-trains the model on only the labeled synthetic data. The main contributions of this paper are below. First, we have established a framework and built its prototype system to annotate privacy-preserving synthetic data in a confidential computing environment. This realizes an annotation process with the privacy protection of real data and models even among collaborators. Second, we have proposed and verified an active learning method for synthetic data that takes advantage of a sandbox environment. This not only improves the efficiency of human work but also reduces the privacy risk by reducing the amount of data disclosure through a synergistic effect of a sandbox architecture and privacy-preserving synthetic data generation.

2 RELATED WORK

2.1 Data Sandbox

Confidential computing is a computing method with information protection in an attested Trusted Execution Environment (TEE) [7]. It has been incorporated into many server central procession units (CPUs) manufactured by, for example, Intel and AMD. TEE and remote attestation are the two main technologies underlying confidential computing. TEE is an isolated execution environment architected such that the memory region is encrypted by a CPU. Remote attestation is a method provided by CPU vendors for users to confirm the authenticity of the TEE.

A data sandbox is a representative implementation of confidential computing. Fig 1 shows the overview of the data sandbox. It enables an analysis using shared data and algorithms confidentially from not only one another but also platform providers [14]. The data sandbox creates a secure container environment on a TEE that is isolated from other computing environments. Remote attestation allows users to verify that the environment is operating properly. Within the container, data and algorithms are sent securely by using a hybrid cryptosystem that uses a pair of secret and public keys generated within the TEE. Then, only result data are retrieved from the container after processing the algorithms.

2.2 Privacy-preserving Data Synthesis

Synthetic data are mainly used as a substitute for sensitive data for external release and utilization. They are generated on the basis of the distribution, statistics, and correlations of all attributes for

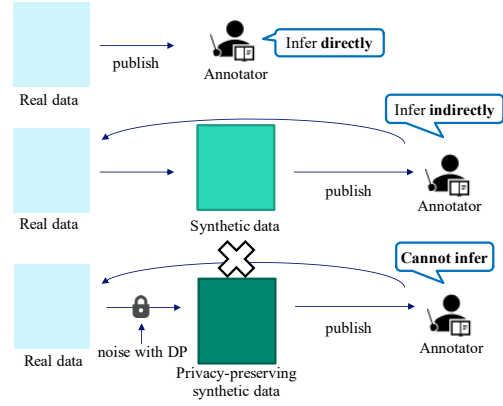


Figure 2: Privacy risks of published data

real data. However, some previous works [4, 5, 25] have pointed out that the information related to the real data is leaked from synthetic data. Specifically, synthetic data generation methods do not guarantee privacy for real data.

To guarantee privacy for real data, DP [10] is often adapted to synthetic data generation. Adding noise for DP to the generative parameters of data synthesis can guarantee privacy for real data. For example, there are several methods such as the statistics based method [1, 18], the graphical model based method [32], and the deep generative model based method [29]. Hereinafter, the data synthesis for adapting DP is called privacy-preserving data synthesis. Fig 2 shows each privacy risk when real data, synthetic data, and privacy-preserving synthesis data are published. Only privacy-preserving synthetic data can protect the privacy of real data from annotators and other users. In addition, privacy-preserving data synthesis can maintain the constant utility for high dimensional data compared with conventional privacy-preserving techniques, i.e., k-anonymization [27] and Pk-anonymization [16]. In this way, privacy-preserving data synthesis is highly able to accelerate sensitive data sharing by achieving both privacy guarantee and data utility.

2.3 Active Learning

Active learning is a representative human-in-the-loop approach toward achieving maximum model performance with minimum annotation cost [24]. The most common setting is pool-based sampling, where a large amount of unlabeled data is available beforehand. In the annotation process, active learning iteratively extracts data that should be annotated for efficient model training, and a human annotates only the extracted data. The expected improvement of model performance is defined as an acquisition function for each data. The main indicators to consider for it are informativeness, representativeness, and diversity. Thus, there are various definitions of acquisition function such as entropy- or Bayesian-approach-based ones for informativeness [11, 31], density- or clustering-based ones for representativeness [15, 30], or a combination of them [8]. In addition, to optimize the iterative process of annotation and model re-training, batch active learning is also proposed. Batch active

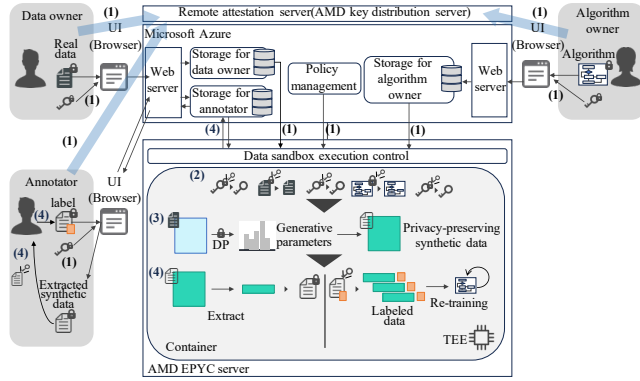


Figure 3: Overview of workflow in Sandbox AI

learning extracts multiple data together considering their combination at each loop for diversity [3, 26].

3 PROTOTYPE SYSTEM AND METHOD

3.1 Sandbox AI Prototype System

In this section, we describe the workflow of the Sandbox AI prototype system. Fig 3 shows the overview of the workflow and the configuration of the prototype system. For easy understanding, each actor is assumed to be a single and independent person. In addition, since it focuses on AI model training, the users of the result data are excluded.

(1) Each information owner generates a symmetric encryption key and encrypts their information with their key. Each owner encrypts the symmetric encryption key with a public key provided by the data sandbox and shares it with the data sandbox. Then, the encrypted data and algorithms are placed in the data sandbox. The owners can verify whether the data sandbox is created under the policy agreed upon in advance, that is, whether it was not swapped with malicious data or algorithms, by referring to the attestation report provided by the system. (2) The data sandbox uses a secret key to decrypt a symmetric key and then decrypt provided information by the symmetric key in the TEE container. (3) Privacy-preserving synthetic data are generated from the placed real data. For easy understanding, the number of generated synthetic data is assumed to be the same as the number of real data. (4) Sandbox AI extracts a few synthetic data which should be annotated, encrypts them, and send them to the annotator. The detail of the extraction method using our proposed active learning is described in the following section. The annotator decrypts them, attaches labels to the synthetic data, encrypts them including the labels, and shares them with the data sandbox. Then, the AI model is re-trained with the updated labeled synthetic data after decrypting the labeled synthetic data. This process (4) is repeated until the model is sufficiently trained. (5) After completion, data and algorithms are deleted along with the data sandbox.

These processes ensure the authenticity of information and keep it secret from other actors. Specifically, during the annotation phase, actors other than the data owner repeatedly interact with the data,

but the use of privacy-preserving synthetic data prevents the information leakage of the real data.

3.2 Active Learning for Sandbox AI

This section describes an active learning method for privacy-preserving synthetic data taking into account a data sandbox architecture to reduce the number of data to be published to the annotator. There is more than one motivation to reduce the number of data to annotate in Sandbox AI. Of course, one advantage is the reduction of annotation cost by humans as well as the use of active learning for supervised learning models in conventional settings. The additional benefit of reducing the number of disclosed synthetic data is to directly reduce the information leakage risk. Even though privacy-preserving synthetic data can achieve both privacy guarantee and data utility, it is inevitable that an over-strict privacy budget setting reduces the data utility. Therefore, appropriate privacy budget settings along with optimization of the amount of disclosed synthetic data lead to further enhancing the advantages of Sandbox AI.

The main metrics for acquisition functions in active learning are informativeness, representativeness, and diversity. It is important to note that the representativeness should be evaluated on the distribution of real data, not on the distribution of synthetic data. This is because the data distribution is shifted due to the noise for DP. Taking the above metrics into account in the shifted distribution does not necessarily improve model performance efficiently. Therefore, this paper proposes a new acquisition function that takes into account the representativeness of the synthetic data in the real data distribution. The proposed function leverages the existence of real data in the TEE container, which are never disclosed to other actors in Sandbox AI.

Let $\mathbf{x}^s \in \mathcal{X}^s$, $\mathcal{X}_s^s$ and $q(\mathcal{X}_s^s)$ denote synthetic data, a small number of data samples selected from $\mathcal{X}^s$, and its acquisition function score, respectively. The sampling by an acquisition function of general batch active learning that considers informativeness and diversity is defined as

$$\hat{\mathcal{X}}_s^s = \operatorname{argmax}_{\mathcal{X}_s^s \subset \mathcal{X}_u^s} q(\mathcal{X}_s^s). \quad (1)$$

The proposed method is based on adding a constraint derived from the concept of a clustering-based method for representativeness. Initially, clustering is performed on the whole real data, $\mathbf{x}^r \in \mathcal{X}^r$, and unlabeled synthetic data, $\mathbf{x}^s \in \mathcal{X}_u^s$. Then, $\mathcal{X}_c^s \subset \mathcal{X}_u^s$ is the set of $\mathbf{x}^s$ whose belonging cluster contains unlabeled synthetic data less than a constant ratio C . If $C = 1$, $\mathcal{X}_c^s$ indicates the set of $\mathbf{x}^s$ that excludes the data that created clusters only with synthetic data. This method samples data from $\mathcal{X}_c^s$ not from whole $\mathcal{X}_u^s$ in accordance with the acquisition function defined as

$$\hat{\mathcal{X}}_s^s = \operatorname{argmax}_{\mathcal{X}_c^s \subset \mathcal{X}_u^s} q(\mathcal{X}_s^s). \quad (2)$$

This additional constraint on the search space of the batch enables data sampling that guarantees DP by privacy-preserving synthetic data as well as representativeness in the real data. This method also has the advantage of computational cost. There is no need to individually compute the distance between the synthetic data and the real data distribution. Note that although real data are

used in the process of calculating the acquisition function in Sandbox AI, the disclosed data are only privacy-preserving synthetic data. This does not lead to any information leakage.

4 EXPERIMENT

We verify that the proposed active learning works on the Sandbox AI prototype system and that it enables efficient annotation.

4.1 Experimental Settings

The dataset for the verification was the Adult Data Set [17] in the UCI Machine Learning Repository [9]. The task was a two-class classification problem where the input consisted of 13 attributes, including continuous and categorical variables such as age and job title, and the output was whether or not the annual income was \$50,000 or more. The total number of valid data after preprocessing to remove missing data was 29,162, and the class-specific numbers were 6,508 and 22,654, respectively.

Sandbox AI has no restrictions on the machine learning models it handles. It is assumed that the model-building process in Sandbox AI is a trial-and-error process using various algorithms. To verify the model agnostic performance, we implemented three classification models: XG-Boost, decision tree, and support vector machine. The performance results were evaluated on the basis of the average performance of the three models.

The synthetic data were generated by a Bayesian network-based method [23]. Its DP was achieved by PrivBayes [32]. The strictness of DP is determined by ϵ in (ϵ, δ) -DP. The smaller ϵ indicates the stricter privacy budget. This paper tested three patterns of DP settings: $\epsilon = 8.0$, which is a common setting in the privacy analysis of machine learning, $\epsilon = 2.0$, which is a stricter setting, and $\epsilon = \infty$, which means that no DP is applied.

The active learning implementation was based on uncertainty-based batch active learning [3] and HDBSCAN clustering [2, 19]. For sampling settings, the number of data to be extracted per loop was set to 10, and the maximum number of loops was set to 200. This means that for each trial under each condition, the total number of annotations was set to 2,000. Evaluation is performed after each annotation and training loop to verify the improvement in model performance with respect to the number of loops. Since both synthetic data generation and sampling by active learning include a random element, we conducted five trials for the same condition and evaluated the results in accordance with the mean value and standard deviation of the acquired results. In addition to the proposed method (Proposed), we also applied the conventional active learning (Conv) [3] without considering real data and random sampling (Random) to verify the efficiency of Proposed. Random sampling is a sampling method in which the acquisition function scores of all unlabeled data are defined as equal constants, so that all data are extracted with equal probability.

The entire dataset was divided into three datasets for the experiment. First, 10% of the total dataset is left as a test dataset to verify the performance of the machine learning models trained in each loop, condition, and trial. Of the remaining 90%, half is used as real data for generating privacy-preserving synthetic data in the TEE container in the Sandbox AI. The other half is used as the training

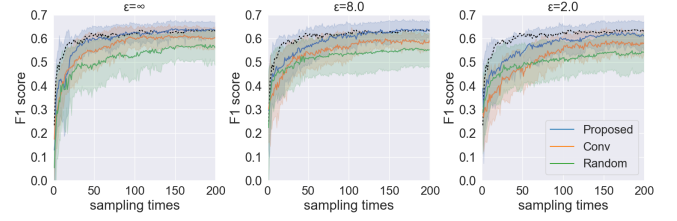


Figure 4: experimental results

dataset for supervised learning with normal real data for the simulated annotator. The main reason for using simulated annotators is to eliminate arbitrary influences by human annotators. Some studies have also shown that this does not cause any problems, e.g. commonly used machine learning datasets contain some label errors [22], and that annotation by machine learning models is similar to that of humans [21]. Note that the explicit division of the training dataset for the simulated annotator model and the real dataset used for generating synthetic data is intended to prevent the information leakage of the original real dataset through the annotation even in the experiment.

4.2 Experimental Results

Fig 4 shows results of classification performance on each loop, sampling method, and privacy budget. The dashed black line shows the result of applying active learning to real data as a reference. In the non-DP setting ($\epsilon = \infty$), Proposed performs as well as the real data results, although it is slower to launch. In the strict DP setting ($\epsilon = 2.0$), the difference from the real data cannot be filled. Interestingly, there is an accuracy inversion between Conv and Random depending on the number of loops in DP settings ($\epsilon = 8.0, 2.0$). This appears to be a result of the data shift caused by the noise of DP and also indicates that conventional active learning is not always effective for privacy-preserving synthetic data. In contrast, Proposed has consistently outperformed the other two methods almost since its launch in all settings. This is a result of taking advantage of the architecture of Sandbox AI and considering representativeness in real data distributions.

5 CONCLUSION

This paper proposed Sandbox AI and a new active learning method for Sandbox AI and demonstrated their superiority in efficient annotation under the constraint of not disclosing real data and model algorithms even among the collaborators. Its superiority is achieved by data sandbox architecture derived from confidential computing, privacy-preserving synthetic data generation, and active learning that takes advantage of the combination of the characteristics of these two technologies. Specifically, the proposed active learning method considers not only the informativeness and diversity of the extracted synthetic data but also their representativeness in the real data distribution thanks to the data sandbox architecture. For future work, we will conduct experiments in medical fields to have actual users experience the system and discuss its practicality including its user interface with domain experts in each field.

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